

EULER-VOIGT EQUATIONS

1. INTRODUCTION

We study the Euler-Voigt equations posed on a periodic domain $\mathbb{T}^3 := (\mathbb{R}/\mathbb{Z})^3$

$$(1.1) \quad \begin{aligned} -\alpha^2 \partial_t \Delta \mathbf{u}_\alpha + \partial_t \mathbf{u}_\alpha + (\mathbf{u} \cdot \nabla) \mathbf{u}_\alpha + \nabla p &= 0, \\ \nabla \cdot \mathbf{u}_\alpha &= 0, \\ \mathbf{u}_\alpha(\mathbf{x}, 0) &= \boldsymbol{\eta}_\alpha, \end{aligned}$$

where $\alpha > 0$ is a regularization parameter. The Euler-Voigt equations can be viewed as an inviscid regularization of the Euler equations

$$(1.2) \quad \begin{aligned} \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p &= 0, \\ \nabla \cdot \mathbf{u} &= 0, \\ \mathbf{u}(\mathbf{x}, 0) &= \boldsymbol{\eta}. \end{aligned}$$

The solutions to (1.2) and (1.1) both possess the following time scaling symmetry

$$(1.3) \quad \tilde{\mathbf{u}}(\mathbf{x}, t) = \lambda \mathbf{u}(\mathbf{x}, \lambda t).$$

Unlike the Euler equations for which the global well-posedness of the solutions remains an open problem, the solutions of the Euler-Voigt are known to be globally well-posed for $t \in (-\infty, \infty)$ [1]. Moreover, the “ α -energy” of the solutions of (1.1) is conserved for all times

$$(1.4) \quad E_\alpha(t) := \|\mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = E_\alpha(0).$$

The behavior of the solutions of (1.1) as $\alpha \rightarrow 0$ can be used to determine whether the solutions of the Euler equations will develop a finite time singularity [2].

Theorem 1.1. *Let $\boldsymbol{\eta}_\alpha \in H^s$, $s \geq 3$ with $\nabla \cdot \boldsymbol{\eta}_\alpha = 0$ and let $\mathbf{u}_\alpha$ be the corresponding unique solution of (1.1). Suppose that there exists $T^* > 0$ such that*

$$(1.5) \quad \limsup_{\alpha \rightarrow 0^+} \left(\alpha \sup_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2} \right) > 0.$$

Then the Euler equations with the initial condition $\boldsymbol{\eta}_\alpha$ must develop a singularity within the interval $[0, T^]$.*

The Theorem 5.1 in [3] stated that

Theorem 1.2. *Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$, let $\mathbf{u}, \mathbf{u}_\alpha \in C([0, T], H^s(\mathbb{T}^3)) \cap C^1([0, T], H^{s-1}(\mathbb{T}^3))$ be the corresponding solutions to (1.2) and (1.1), respectively, where $0 < T < T_{\max}$ and $T_{\max}$ is the maximal time for which a solution to (1.2) exists and is unique. For all $t \in [0, T]$, we have*

$$(1.6) \quad \begin{aligned} &\|\mathbf{u}(t) - \mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla(\mathbf{u}(t) - \mathbf{u}_\alpha(t))\|_{L^2}^2 \\ &\leq \left(\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \right) e^{Ct} + C\alpha^2(e^{Ct} - 1). \end{aligned}$$

Based on Theorem 1.2, we know that if

$$(1.7) \quad \|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \rightarrow 0, \quad \text{as } \alpha \rightarrow 0,$$

then for all $t \in [0, T]$,

$$(1.8) \quad \|\mathbf{u}(t) - \mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla(\mathbf{u}(t) - \mathbf{u}_\alpha(t))\|_{L^2}^2 \rightarrow 0.$$

Therefore, we can put the constraint on $\|\nabla \boldsymbol{\eta}_\alpha\|_{L^2}$ in the optimization process and check whether $\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 \rightarrow 0$ as $\alpha \rightarrow 0$ in computation.

We slightly modify Theorem 5.2 in [3] as follows.

Theorem 1.3. *Given initial data $\eta, \eta_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$ satisfying*

$$(1.9) \quad \|\eta - \eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\eta - \eta_\alpha)\|_{L^2}^2 \rightarrow 0, \quad \text{as } \alpha \rightarrow 0,$$

let $\mathbf{u}, \mathbf{u}_\alpha \in C([0, T], H^s(\mathbb{T}^3)) \cap C^1([0, T], H^{s-1}(\mathbb{T}^3))$ be the corresponding solutions to (1.2) and (1.1), respectively.

If there exists a finite time $T^ > 0$ such that the solutions $\mathbf{u}_\alpha$ satisfies*

$$(1.10) \quad \limsup_{\alpha \rightarrow 0} \left(\alpha^2 \sup_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 \right) > 0,$$

then $\mathbf{u}$ develops a singularity in the interval $[0, T^]$.*

Proof. Suppose that the solution $\mathbf{u}$ of the Euler equations (1.2) with the initial condition η is regular on $[0, T^*]$. Since the Euler-Voigt equations are globally well-posed, we have

$$(1.11) \quad \|\mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \eta_\alpha\|_{L^2}^2,$$

thus

$$(1.12) \quad \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \eta_\alpha\|_{L^2}^2 - \|\mathbf{u}_\alpha(t)\|_{L^2}^2.$$

Taking the supremum over $t \in [0, T^*]$ of the above equation, we obtain that

$$(1.13) \quad \sup_{t \in [0, T^*]} \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \eta_\alpha\|_{L^2}^2 - \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2}^2.$$

Taking the lim sup when $\alpha \rightarrow 0$ of the above equation, we obtain that

$$(1.14) \quad \begin{aligned} \limsup_{\alpha \rightarrow 0} \sup_{t \in [0, T^*]} \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 &= \limsup_{\alpha \rightarrow 0} \left(\|\eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \eta_\alpha\|_{L^2}^2 - \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2}^2 \right) \\ &= \|\eta\|_{L^2}^2 - \|\eta\|_{L^2}^2 \\ &= 0, \end{aligned}$$

which contradicts the hypothesis in the theorem that the above expression is positive. The second equality holds because

$$\begin{aligned} \limsup_{\alpha \rightarrow 0} \left| \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2} - \|\eta\|_{L^2} \right| &\leq \limsup_{\alpha \rightarrow 0} \|\mathbf{u}_\alpha(t_\alpha) - \eta\|_{L^2}^2 \\ &\leq \limsup_{\alpha \rightarrow 0} \|\mathbf{u}_\alpha(t_\alpha) - \mathbf{u}(t_\alpha)\|_{L^2}^2 + \|\mathbf{u}(t_\alpha) - \eta\|_{L^2}^2 \\ &= 0. \end{aligned}$$

Here in the second inequality, we use the fact that the convergence of the solutions of the Euler-Voigt equations to the solution of the Euler equations is uniform on $[0, T^*]$, as shown in Theorem 1.2. □

2. PDE OPTIMIZATION METHOD

2.1. The optimization problem. We consider analytic initial conditions that belong to the following Gevrey class

$$(2.1) \quad G^\sigma := \left\{ \mathbf{v} \in H^s(\mathbb{T}^3) : \|\mathbf{v}\|_{G^\sigma} = \langle \mathbf{v}, \mathbf{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0,$$

with the inner product

$$(2.2) \quad \begin{aligned} \forall \mathbf{v}, \mathbf{u} \in G^\sigma, \quad \langle \mathbf{v}, \mathbf{u} \rangle_{G^\sigma} &:= \sum_{j \in \mathbb{Z}^3} (1 + |2\pi j|^2)^s e^{4\pi\sigma|j|} \hat{\mathbf{v}}_j \cdot \bar{\hat{\mathbf{u}}}_j \\ &= \int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v} \cdot \mathbf{u} \, dx, \end{aligned}$$

where overbar denotes complex conjugation, whereas the operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$(2.3) \quad \left[\widehat{|D|v} \right]_j := 2\pi |j| \hat{v}_j, \quad \left[\widehat{e^{\sigma|D|}v} \right]_j := e^{2\pi\sigma|j|} \hat{v}_j.$$

Since the initial conditions in (1.1) and (1.2) need to be divergence-free and the quantity $\int_{\mathbb{T}} \eta \, dx$ is an invariant of motion, we introduce the following subspace in which optimal initial conditions will be sought

$$(2.4) \quad V := \{v \in G^\sigma : \nabla \cdot v = 0, \quad \int_{\mathbb{T}^3} v \, dx = \mathbf{0}\}.$$

According to the analytical results in Section 1, we define the objective functional $\Phi_{\alpha,T} : V \rightarrow \mathbb{R}^+$ as

$$(2.5) \quad \Phi_{\alpha,T}(\eta) := \|\nabla u_\alpha(T; \eta)\|_{L^2}^2 = \|u_\alpha(T; \eta)\|_{\dot{H}^1}^2,$$

where $u_\alpha(T; \eta)$ is the solution of the Euler-Voigt equations (1.1) obtained with the initial condition $\eta \in V$ and α is the regularization parameter in (1.1).

Due to the scaling property (1.3) of the Euler-Voigt equations (1.1), we can restrict our discussion to initial conditions with unit $\dot{H}^1$ seminorm which belong to a closed manifold $\mathcal{M}_1 \subset V$ defined as

$$(2.6) \quad \mathcal{M}_1 = \{v \in V : \|v\|_{\dot{H}^1} = 1\}.$$

Therefore, we arrive at the following optimization problem

Problem 2.1. Given $\alpha, T \in \mathbb{R}_+$, find

$$(2.7) \quad \tilde{\eta}_{\alpha,T} = \operatorname{argmax}_{\eta \in \mathcal{M}_1} \Phi_{\alpha,T}(\eta).$$

Our goal is to first fix T and solve the optimization problem 2.1 for a sequence of decreasing values of α that converge to 0.

2.2. The adjoint method. We compute the gradient $\nabla \Phi_{\alpha,T}(\eta)$ of the objective functional in (2.5) using an adjoint method which depends on solving a properly defined adjoint system backward in time. We first introduce the Gâteaux (directional) differential $\Phi'_{\alpha,T}(\eta; \eta') : V \times V \rightarrow \mathbb{R}$

$$(2.8) \quad \begin{aligned} \Phi'_{\alpha,T}(\eta; \eta') &:= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\Phi_{\alpha,T}(\eta + \epsilon \eta') - \Phi_{\alpha,T}(\eta)] \\ &= 2 \langle u_\alpha(\cdot, T), u'_\alpha(\cdot, T) \rangle_{\dot{H}^1}. \end{aligned}$$

Here $u'_\alpha(x, t)$ is the solution of the linearization of the Euler-Voigt equations (1.1) around its solution $u_\alpha(x, t; \eta)$, which is defined by

$$(2.9) \quad \begin{aligned} \mathcal{L} \begin{bmatrix} u'_\alpha \\ p'_\alpha \end{bmatrix} &:= \begin{bmatrix} \partial_t u'_\alpha + (I - \alpha^2 \Delta)^{-1} (u'_\alpha \cdot \nabla u_\alpha + u_\alpha \cdot \nabla u'_\alpha + \nabla p'_\alpha) \\ \nabla \cdot u'_\alpha \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix}, \\ u'_\alpha(x, 0) &= \eta'(x), \end{aligned}$$

where η' is the perturbation of the initial condition and $p'_\alpha(x, t)$ is the corresponding pressure perturbation. We further introduce the adjoint states $\{u_\alpha^* : \mathbb{T}^3 \times [0, T] \rightarrow \mathbb{R}^3, p_\alpha^* : \mathbb{T}^3 \times [0, T] \rightarrow \mathbb{R}\}$

and the following duality-pairing relation

$$\begin{aligned}
 \left(\mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix}, \begin{bmatrix} \mathbf{u}^*_\alpha \\ p^*_\alpha \end{bmatrix} \right) &:= \int_0^T \int_{\mathbb{T}^3} \mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix} \cdot \begin{bmatrix} \mathbf{u}^*_\alpha \\ p^*_\alpha \end{bmatrix} dx dt \\
 (2.10) \quad &= \left(\begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix}, \mathcal{L}^* \begin{bmatrix} \mathbf{u}^*_\alpha \\ p^*_\alpha \end{bmatrix} \right) + \int_{\mathbb{T}^3} \mathbf{u}'_\alpha(\mathbf{x}, T) \cdot \mathbf{u}^*_\alpha(\mathbf{x}, T) dx \\
 &\quad - \int_{\mathbb{T}^3} \boldsymbol{\eta}'(\mathbf{x}) \cdot \mathbf{u}^*_\alpha(\mathbf{x}, 0) dx \\
 &= 0.
 \end{aligned}$$

Here “ $\cdot$ ” denotes the usual Euclidean inner product on $\mathbb{R}^4$ and $\mathcal{L}^*$ is the *adjoint* operator defined in terms of the following system, which is obtained by performing integration by parts with respect to both space and time in (2.10)

$$\begin{aligned}
 (2.11) \quad \mathcal{L}^* \begin{bmatrix} \mathbf{u}^*_\alpha \\ p^*_\alpha \end{bmatrix} &:= \begin{bmatrix} -\partial_t \mathbf{u}^*_\alpha - (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}^*_\alpha + (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}^*_\alpha)^T) \mathbf{u}_\alpha - \nabla p^*_\alpha \\ -\nabla \cdot \mathbf{u}^*_\alpha \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix}, \\
 \mathbf{u}^*_\alpha(\mathbf{x}, T) &= 2|D|^2 \mathbf{u}_\alpha(\mathbf{x}, T).
 \end{aligned}$$

Therefore, we have

$$(2.12) \quad \Phi'_{\alpha, T}(\boldsymbol{\eta}; \boldsymbol{\eta}') = 2 \langle \mathbf{u}_\alpha(\cdot, T), \mathbf{u}'_\alpha(\cdot, T) \rangle_{\dot{H}^1} = \langle \mathbf{u}^*_\alpha(\cdot, 0), \boldsymbol{\eta}'(\cdot) \rangle_{L^2}$$

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